Semester Project – Parallel and Distributed Computing (PDC)

Project Title: High-Rate Network Traffic Analyzer for Early DDoS Detection and Mitigation
Group Size: Three students max
Submission Format: Report + Code

What to deliver

Each group must submit:

1. **Source Code** (well-documented and executable).
2. **Experiment Scripts** (for replay, testing, and metrics collection).
3. **Performance Report** (10–12 pages, including all analysis).
4. **Presentation Slides** (for viva/demo).

Project Overview

Distributed Denial of Service (DDoS) attacks pose one of the most significant challenges to network security. Detecting and mitigating these attacks in real time requires high-performance computation and efficient parallel/distributed system design.

In this project, your team will **design and implement a high-rate network traffic analyzer** capable of **detecting and blocking DDoS attacks** using multiple detection algorithms implemented on a **parallel platform**.
You will analyze detection performance, blocking efficiency, and scalability.

Each group must **select and implement ONE of the following scenarios:**

- **Scenario A:** GPU-based implementation using **OPENCL**
- **Scenario B:** Cluster-based implementation using **MPI**

Project Objectives

1. Develop a high-performance DDoS detection system using parallel/distributed programming techniques.
2. Implement **at least two DDoS detection algorithms** (from literature) on network flow or packet data.
3. Implement **at least 1 blocking/mitigation methods** for detected DDoS traffic.
4. Evaluate detection accuracy, latency, throughput, and scalability.

5. Analyze and compare performance results across rates, algorithms, and configurations.

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3. Link to OBE and Complex Computing Problem (CCP)

This semester project directly addresses a **Complex Computing Problem (CCP)** as required by the **Outcome-Based Education (OBE)** framework of the department and **NCEAC accreditation guidelines**.

- **Nature of Complexity:**
  The problem involves the design and analysis of a **parallel and distributed system** for real-time network security. It requires integrating concepts of **parallel programming**, **network traffic analysis**, and **cyber defense mechanisms**—demanding the use of advanced computing knowledge and specialized techniques beyond standard computing practices.
- **Justification as a CCP:**
  - The problem has **no known deterministic solution** and requires **research-based decision-making** (choice of algorithms, data partitioning, GPU kernel design, etc.).
  - It involves **multiple interacting components** (data ingestion, detection, blocking, and performance evaluation).
  - The system must handle **real-world complexity**, including large-scale datasets and high-speed network traffic.
  - Students must consider **performance trade-offs** (accuracy vs. speed, detection latency vs. resource usage).
  - It aligns with **Program Learning Outcomes (PLO 3 & PLO 4)** — problem analysis and design/development of solutions for complex computing problems.

Students must explicitly discuss **why this problem qualifies as a CCP** in the **introduction section of their report**, including:

- Complexity in computation and system design
- Multiple solution approaches considered
- Justification of the chosen parallel/distributed method
- Evaluation of solution performance and limitations

Scenarios(Attempt 1 of 2)

Scenario A – GPU-Based Analyzer (OPENCL Implementation)

Goal:
Implement a high-throughput DDoS detection system using **OPENCL kernels** to parallelize heavy computations such as entropy calculation, feature extraction, or ML-based inference.

Requirements:

1. Input traffic: Use **CIC-DDoS2019** dataset (preferred) or **CAIDA DDoS 2007**.
2. Implement at least **two detection algorithms**, for example:
    - Entropy-based detection
    - CUSUM or PCA-based statistical detection
    - Machine Learning (Random Forest / SVM / Lightweight NN)
3. Offload at least one major computation to **GPU kernels** using OPENCL.
4. Implement **two blocking methods**, such as:
    - Remote Triggered Black Hole (RTBH)
    - Access Control List (iptables) simulation
    - BGP FlowSpec simulation
5. Measure and report:
    - Detection latency (how early DDoS detected)
    - Accuracy metrics (TPR, FPR, Precision, Recall, F1)
    - Throughput (packets/sec or Gbps)
    - GPU utilization and kernel performance
    - Blocking efficiency (how much attack traffic was blocked vs legitimate)

Deliverables:

- Source code (OPENCL + host code)
- GPU kernel files and documentation
- Experiment logs and plots
- Final report

Scenario B – Cluster-Based Analyzer (MPI Implementation)

Goal:
Develop a distributed DDoS detection and blocking framework using **MPI** on a simulated or real cluster (3–8 nodes).

Requirements:

1. Input traffic: Use **CIC-DDoS2019** dataset (preferred) or **CAIDA DDoS 2007**.

2. Partition traffic or flow data among MPI nodes (hash-based, time-based, or flow-based partitioning).
3. Implement at least **three detection algorithms**, for example:
    - Entropy-based
    - PCA or CUSUM statistical detection
    - Machine Learning or time-series based
4. Use MPI processes for:
    - Parallel analysis per node
    - Alert aggregation and decision at a coordinator node
5. Implement **two blocking mechanisms**, such as:
    - RTBH simulation
    - FlowSpec rule generation
    - Rate-limiting or ACL rules
6. Measure and report:
    - Detection latency (local and global)
    - MPI communication overhead
    - Accuracy and false alarm rate
    - Throughput (scaling with number of nodes)
    - Blocking effectiveness and collateral damage

Deliverables:

- Source code (MPI/C/C++/Python)
- Scripts for running distributed experiments
- Cluster configuration and test setup details
- Final report and demo video

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5. Recommended Datasets

1. **CIC-DDoS2019** – Canadian Institute for Cybersecurity Dataset
2. **CAIDA DDoS Attack 2007** – CAIDA Dataset
3. (Optional) **MAWI Traffic Archive** – for mixed background traffic.

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6. Suggested DDoS Detection Algorithms (choose any 2)

1. Statistical

2. ML-Based
3. Deep Learning

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7. Recommended Blocking / Mitigation Methods

Method

1. Remote Triggered Black Hole (RTBH)
2. BGP FlowSpec
3. Rate Limiting
4. ACL / iptables
5. SDN Controller

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8. Performance Analysis Requirements(must ensure all these)

Your report must include quantitative evaluation of the following metrics:

1. **Detection Lead Time (ms)** – time between attack start and first alert.
2. **Accuracy Metrics** – Precision, Recall, F1, False Positive Rate.
3. **Throughput** – packets/sec and Gbps processed.
4. **Latency** – average and 95th percentile packet processing time.
5. **Resource Utilization** – CPU/GPU usage, memory, and network.
6. **Blocking Effectiveness** – attack traffic dropped (%) and collateral impact (%).
7. **Scalability Analysis** – performance vs. increased input rate or cluster size.

Graphs and tables must be provided for all metrics.

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12. Tools and Libraries

- **Programming Languages:** C
- **Parallel Frameworks:** OPENCL / MPI
- **Packet Tools:** tcpreplay, tshark, scapy, Zeek
- **Analysis:** pandas, numpy, matplotlib, Wireshark
- **Blocking Simulation:** iptables, tc, or custom scripts

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13. Expected Learning Outcomes

By completing this project, students will:

- Understand the design of scalable and parallel DDoS detection systems.
- Gain hands-on experience in **OPENCL** or **MPI** for parallel/distributed computing.
- Learn to analyze real network datasets and measure system performance.
- Apply cybersecurity concepts (detection + response) to real-time systems.
- Produce an end-to-end system prototype simulating a **real-world DDoS defense pipeline**.
- Demonstrate competence in **solving Complex Computing Problems (CCPs)** as required by the **OBE framework**.